Experiment 3: Designing a calculator using Typescript

Program:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Simple Calculator</title>
  <style>
    /* Base styles to center everything and set a nice background */
    body {
      font-family: 'Segoe UI', Roboto, Helvetica, Arial, sans-serif;
      background-color: #f3f4f6; /* Light gray background */
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }

    /* The main calculator card */
    .calculator {
      background-color: #ffffff;
      padding: 2rem;
      border-radius: 12px;
      box-shadow: 0 10px 25px rgba(0, 0, 0, 0.1);
      width: 100%;
      max-width: 320px;
    }

    h2 {
      text-align: center;
      color: #1f2937;
      margin-top: 0;
      margin-bottom: 1.5rem;
    }

    /* Styling the inputs to look modern and clean */
    .input-group {
      display: flex;
      flex-direction: column;
```

```
    gap: 12px;
    margin-bottom: 20px;
}

.input-group input {
    padding: 12px 15px;
    border: 1px solid #d1d5db;
    border-radius: 8px;
    font-size: 16px;
    transition: border-color 0.2s, box-shadow 0.2s;
    outline: none;
}

/* Blue glow when clicking on an input */
.input-group input:focus {
    border-color: #3b82f6;
    box-shadow: 0 0 0 3px rgba(59, 130, 246, 0.2);
}

/* CSS Grid to evenly space the 4 operator buttons */
.button-group {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 10px;
    margin-bottom: 20px;
}

.button-group button {
    background-color: #3b82f6;
    color: white;
    border: none;
    padding: 12px;
    border-radius: 8px;
    font-size: 18px;
    font-weight: bold;
    cursor: pointer;
    transition: background-color 0.2s, transform 0.1s;
}

/* Hover and click effects for buttons */
.button-group button:hover {
    background-color: #2563eb;
}

.button-group button:active {
    transform: scale(0.95);
}
```

```
      /* A distinct area for the result to stand out */
      .result-box {
         background-color: #f3f4f6;
         padding: 15px;
         border-radius: 8px;
         text-align: center;
         border: 1px dashed #d1d5db;
      }

      #result {
         margin: 0;
         color: #111827;
         font-size: 1.25rem;
         font-weight: 600;
      }
   </style>
</head>
<body>

   <div class="calculator">
      <h2>Calculator</h2>

      <div class="input-group">
         <input type="number" id="num1" placeholder="Enter first number">
         <input type="number" id="num2" placeholder="Enter second number">
      </div>

      <div class="button-group">
         <button type="button" onclick="calci('+')">+</button>
         <button type="button" onclick="calci('-')">-</button>
         <button type="button" onclick="calci('*')">×</button>
         <button type="button" onclick="calci('/')">÷</button>
      </div>

      <div class="result-box">
         <h3 id="result">Result</h3>
      </div>
   </div>

   <!-- Make sure your script.js is in the same folder -->
   <script src="script.js"></script>
</body>
</html>
```

script.ts

```
function calci(op: string): void {
    // 1. Grab the elements once to improve performance
    const num1Input = document.getElementById("num1") as HTMLInputElement;
    const num2Input = document.getElementById("num2") as HTMLInputElement;
    const resultElement = document.getElementById("result") as HTMLHeadingElement;

    // 2. Validate that inputs are not empty before converting to numbers
    if (num1Input.value.trim() === "" || num2Input.value.trim() === "") {
        resultElement.textContent = "Please enter both numbers";
        resultElement.style.color = "#ef4444"; // Error red
        return;
    }

    let num1: number = Number(num1Input.value);
    let num2: number = Number(num2Input.value);
    let result: number = 0;

    // Reset color to normal in case there was a previous error
    resultElement.style.color = "#111827";

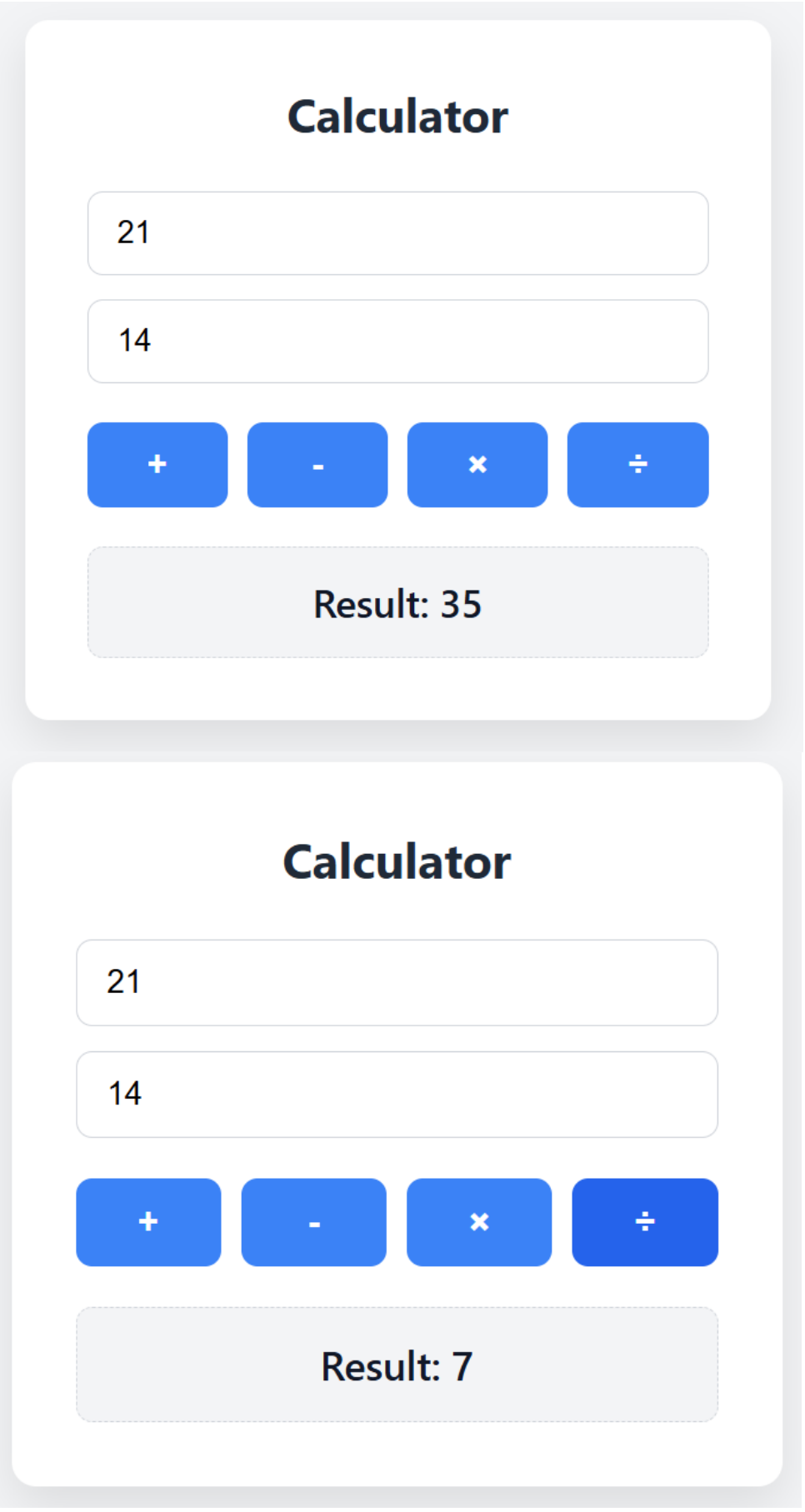
    // 3. Perform the calculation
    switch(op) {
        case '+':
            result = num1 + num2;
            break;
        case '-':
            result = num1 - num2;
            break;
        case '*':
            result = num1 * num2;
            break;
        case '/':
            if (num2 === 0) {
                resultElement.textContent = "Cannot divide by zero";
                resultElement.style.color = "#ef4444";
                return;
            }
            result = num1 / num2;
            break;
        default:
            result = 0;
    }

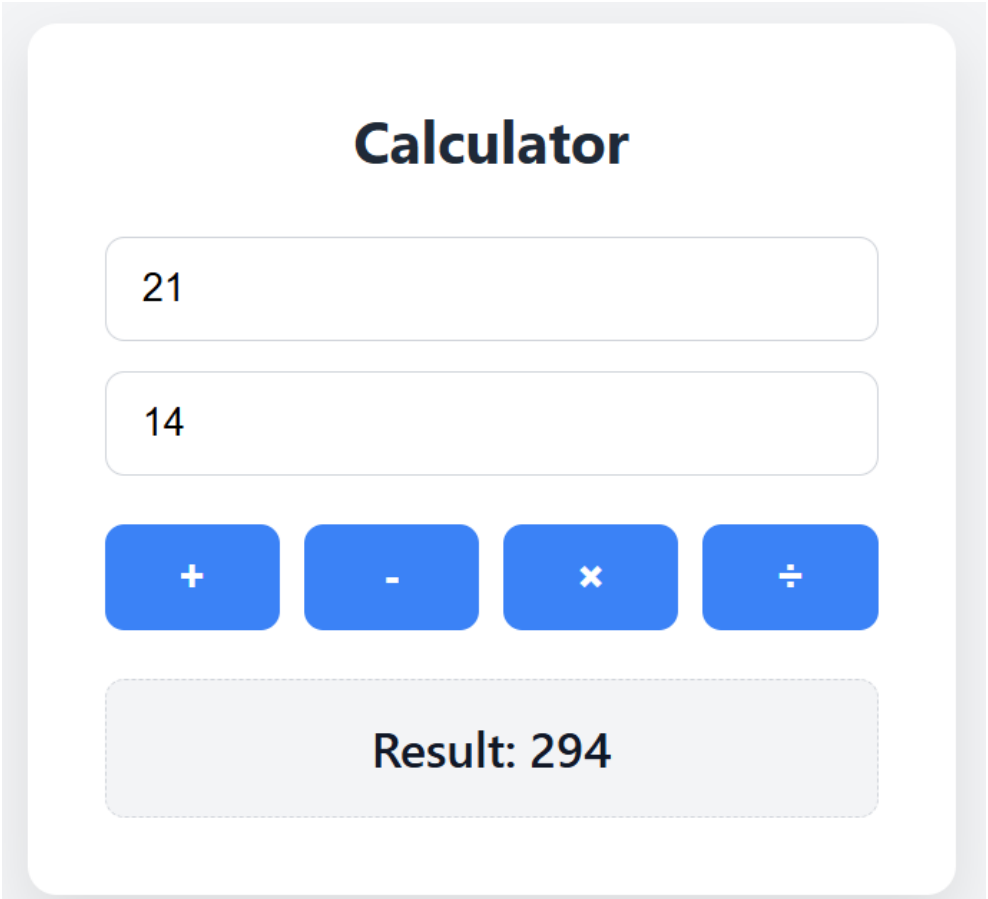
    // 4. Update the DOM exactly once
    resultElement.textContent = "Result: " + result;
}
```

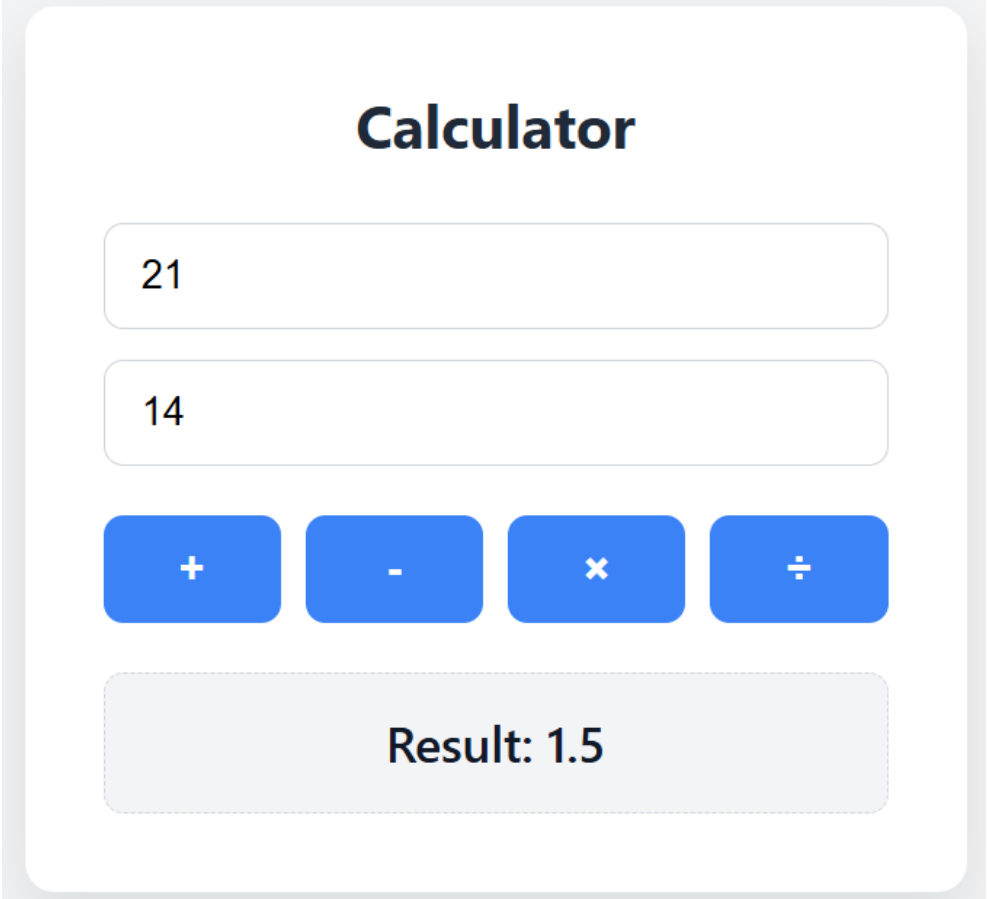
script.js

```
"use strict";
function calci(op) {
   // 1. Grab the elements once to improve performance
   const num1Input = document.getElementById("num1");
   const num2Input = document.getElementById("num2");
   const resultElement = document.getElementById("result");
   // 2. Validate that inputs are not empty before converting to numbers
   if (num1Input.value.trim() === "" || num2Input.value.trim() === "") {
      resultElement.textContent = "Please enter both numbers";
      resultElement.style.color = "#ef4444"; // Error red
      return;
   }
   let num1 = Number(num1Input.value);
   let num2 = Number(num2Input.value);
   let result = 0;
   // Reset color to normal in case there was a previous error
   resultElement.style.color = "#111827";
   // 3. Perform the calculation
   switch (op) {
      case '+':
         result = num1 + num2;
         break;
      case '-':
         result = num1 - num2;
         break;
      case '*':
         result = num1 * num2;
         break;
      case '/':
         // Prevent division by zero
         if (num2 === 0) {
            resultElement.textContent = "Cannot divide by zero";
            resultElement.style.color = "#ef4444";
            return;
         }
         result = num1 / num2;
         break;
      default:
         result = 0;
   }
   // 4. Update the DOM exactly once
   resultElement.textContent = "Result: " + result;
}
```

Output:




Calculator
21
14
+
-
×
÷
Result: 294


Calculator
21
14
+
-
×
÷
Result: 1.5

Calculator

21

0

+ - × ÷

Cannot divide by zero